

Lab Report No 8
Demonstration and practice of 7-Segment Display
Control and display different patterns
Embedded Systems



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Lab Task No 3:

Solution:

Brief description (3-5 lines)
It is aimed to display a series of numbers continually and sequentially counting backward from 90 to 09 on DS3 and advancing from 09 to 90 on DS4. The concept here is simply having two counters: one for decreasing and the other for increasing; a half-second display-time between the showing is kept to keep things in sync. The logic flow involves beyond setting the counters at bounds, a continual cycle is maintained throughout.
The code
ORG 000H MOV A,#90H MOV B,#09H NEXT: MOV P2,A MOV P3,B CALL DELAY DEC A INC B MOV C,A CJNE C,#08H,NEXT MOV A,#90H MOV C,B CJNE C,#91H,NEXT MOV B,#09H JMP NEXT DELAY: MOV R5,#5 DL2: MOV R6,#200 DL1: MOV R7,#249

```
DJNZ R7,$  
DJNZ R6,DL1  
DJNZ R5,DL2  
RET  
END
```

The results (Screenshot)

Build Output

```
creating hex file from ".\Objects\sevenssegmentcounter"...\n".\Objects\sevenssegmentcounter" - 0 Error(s), 3 Warning(s).\nBuild Time Elapsed: 00:00:00
```





Conclusion

In conclusion, The comprehensive exploration of a 7-segment display exposed the difficulties of digital display technology, which can control the segments to depict specific numeric and alphanumeric representations. We learned segment mappings, patterns generation, and real-time display handling for showcasing the capabilities of this simple electronic component. Main learning outcomes included bit manipulation, efficient control algorithms, and hands-on practical application techniques across myriad electronic applications. Such a demonstration brought the basic principles of digital outputs very close to hand experiencing away from theoretical knowledge into new practical hands-on skills in embedded system design. Finally, it became this theoretical foundation tangible learning experience in regard to digital display technology's basic principles.